InsightGlobal

HEALTHCARE AI & AGENTIC SOLUTIONS

Five knowledge-graph platforms across the healthcare value chain

IGH PRESIDENT BRIEFING · NOVEMBER 2026

Agenda

1. Who we are
2. Our approach
3. **CareGraph** · Care Engagement
4. **TherapyGraph** · Life Sciences
5. **RxGraph** · Pharmacy & Drug
6. **DeviceGraph** · Medical Devices & IoT
7. **ChainGraph** · Manufacturing & Supply Chain
8. How we engage

A core team built for any healthcare requirement

From strategy to production across AI, data, Salesforce and cloud, with healthcare compliance built in.

Agentic AI & GenAI engineering

Multi-agent workflows, RAG / GraphRAG, evaluation and guardrails, model-agnostic.

Knowledge graph & data engineering

Healthcare ontologies (RxNorm, ICD-10, SNOMED, LOINC, NPI, UDI), ETL, vector search.

Salesforce for healthcare

Health Cloud, Life Sciences Cloud, Marketing Cloud, Data Cloud, Agentforce, MuleSoft.

Multi-cloud delivery

Google Cloud, AWS and Azure. The same design runs wherever the client already is.

Healthcare compliance & governance

HIPAA, GxP / Part 11, FDA device and safety rules, consent, masking and audit trails.

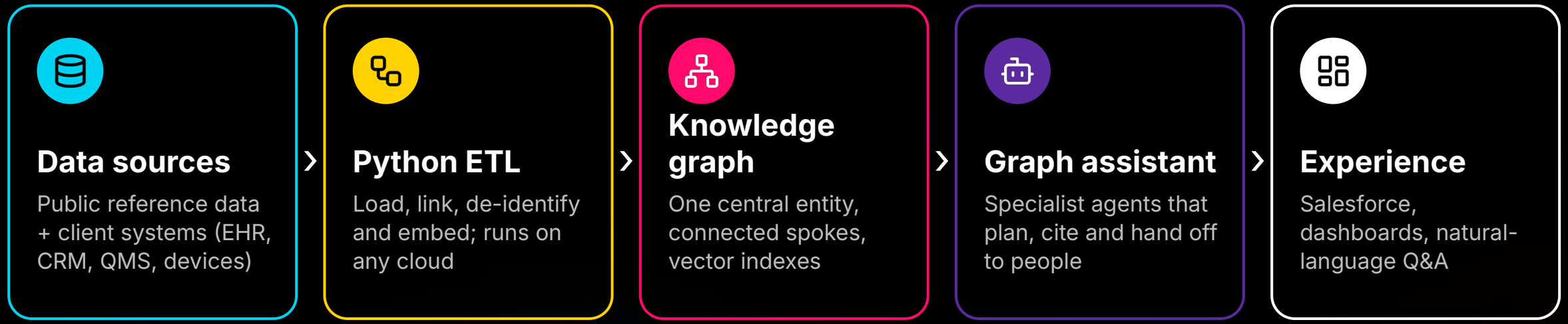
Product & delivery

Ideation workshops, architecture, proof of concept to production, change management.

PROOF POINT We built a governed agentic delivery platform for a large US health system (Salesforce Health and Marketing Cloud, masked data context, compliance guardian, human approval gates).

One pattern, five graphs

Every platform: public + client data → Python ETL → knowledge graph → agents → the tools people already use



Governance built in: guardrails in and out · PHI masking & minimum-data access · human approval gates · evaluation sets · audit trail

Runs on any cloud — Google Cloud · AWS · Azure · Salesforce

Five graphs across the healthcare value chain

Each graph is built around one central entity and solves the whole set of problems in its domain.



CareGraph

CARE ENGAGEMENT

Center: **Patient**

- Cohort & gap intelligence
- Journey designer
- Access navigator
- Referral integrity
- Compliance guardian



TherapyGraph

LIFE SCIENCES

Center: **Therapy**

- HCP 360 & next-best-action
- MLR pre-review
- Trial feasibility & sites
- Patient matching & recruitment
- Medical-affairs insight



RxGraph

PHARMACY & DRUG

Center: **Medication**

- Medication reconciliation
- Interaction & PGx safety
- Benefits & prior auth
- Adherence & adverse events
- Shortage substitution



DeviceGraph

MEDICAL DEVICES & IOT

Center: **Device (UDI)**

- Remote-monitoring triage
- Complaint & reportability
- Recall & UDI tracing
- Post-market trends
- Device cybersecurity



ChainGraph

MANUFACTURING & SUPPLY CHAIN

Center: **Product lot**

- Supplier & material risk
- Deviation & CAPA copilot
- Batch-record review
- Inspection readiness
- Shortage early warning

THE PROBLEM

Care teams can't see, or act on, the whole patient journey

Care gaps, scheduling, referrals, costs and outreach sit in separate systems. Health systems run many campaigns but can't tell why a gap exists, what works for whom, or whether outreach is fair.

Who feels it: Population health, patient access, care coordination, marketing, quality

PROBLEMS IT SOLVES

Care gaps stay open
Screenings and chronic-care follow-ups are missed

One-size-fits-all outreach
Low conversion and high opt-outs

Hard to get an appointment
Long queues and abandoned scheduling

Referral leakage
Patients leave the network

Cost confusion
No clear estimates or financial help

Privacy slows everything
Each campaign waits on compliance review

OUR SOLUTION

CareGraph

Care Engagement Intelligence Platform

One knowledge graph of the patient journey, with agents that find gaps, design outreach and book care, using masked data.

- **Cohort & gap intelligence** explains every gap by measure, community and access
- **Journey designer** personalized Marketing Cloud journeys
- **Access navigator** right provider, right slot, clear estimate
- **Referral integrity** keeps referrals in network
- **Compliance guardian** consent, HIPAA and TCPA checked before send

BUILT ON

Health Cloud | Marketing Cloud | Data Cloud | Agentforce | Tableau

CareGraph / Care Engagement Intelligence Platform

Connecting patients, care gaps, providers, access and outreach with a knowledge graph

Data sources

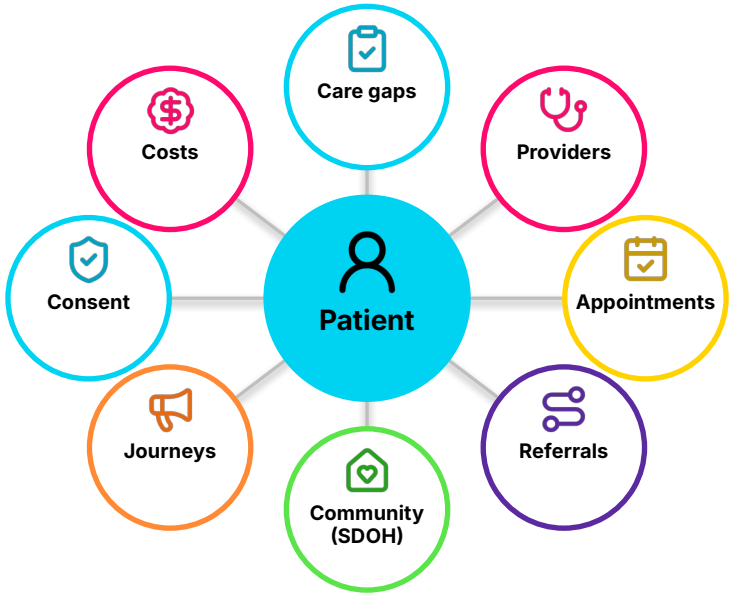
- **EHR / FHIR** — Encounters, problems, care gaps
- **CMS quality measures** — eCQM specifications
- **CDC PLACES + SDOH** — Community health by census tract
- **NPPES** — Provider identity (NPI)
- **Price transparency** — Hospital & payer rate files
- **Engagement data** — Marketing Cloud, contact center

Python ETL

- load_fhir
  load_ecqm
- load_places
  load_nppes
- ingest_rates
  ingest_engagement
- mask_phi
  embed_nodes

Knowledge graph

10 Entity types · **18** Relationships · **6** Vector indexes



Read & intelligence layer

Graph API
REST · GraphQL · MCP

Graph Assistant
agents · any LLM

- Cohort builder
- Gap explainer
- Journey designer
- Access navigator
- Compliance guardian

PHI masking · guardrails in & out · human approval · audit trail

- **Health Cloud console**
- **Marketing Cloud journeys**
- **Natural-language Q&A**

Key capabilities

Care-gap closure · Personalized journeys · Patient access · Referral integrity · Equity analytics · GraphRAG Q&A · Agentic AI

VALUE & BENEFITS

Close more care gaps, book more visits and keep patients in network, with every send compliant.

↑

Care-gap closure rate per campaign

Pilot target

↓

Calls per booked appointment

Pilot target

29%

faster case resolution reported with Health Cloud

Source: Salesforce (vendor-reported)

100%

of outreach checked for consent and PHI before send

Design guarantee

Business

Quality-program revenue, more booked visits, less leakage

Clinical

Earlier screening and chronic-care follow-up

IT & data

One governed patient graph reused by every team

Compliance

Minimum-data access; every send auditable

THE PROBLEM

Commercial, medical and clinical teams work from different pictures

HCP data, trial data, publications and approved content live in separate systems. Brands struggle to personalize compliantly, MLR review takes weeks, and trials miss enrollment targets.

Who feels it: Brand and omnichannel teams, Medical Affairs, MLR, Clinical Operations

PROBLEMS IT SOLVES

Fragmented HCP view
CRM, claims and public data never meet

Slow MLR review
Every asset waits weeks for approval

Off-label risk
Claims drift from the approved label

Trial enrollment delays
Criteria are free text; screening is manual

Site selection by gut feel
Not grounded in patient data

Field insights get lost
MSL notes never reach strategy

OUR SOLUTION

TherapyGraph

Life Sciences Intelligence Platform

One graph of each therapy's world (HCPs, evidence, trials and content) with agents for engagement, review and recruitment.

- **HCP 360 & next-best-action** channel, content and timing per HCP
- **MLR pre-review** flags off-label claims before human review
- **Trial feasibility & sites** masked patient counts by site and region
- **Patient matching & recruitment** every criterion explained, consent-aware outreach
- **Medical-affairs insight** summarizes field and literature signals

BUILT ON

Life Sciences Cloud | Marketing Cloud | Data Cloud | Agentforce

TherapyGraph / Life Sciences Intelligence Platform

Connecting therapies, HCPs, trials, evidence and content with a knowledge graph

Data sources

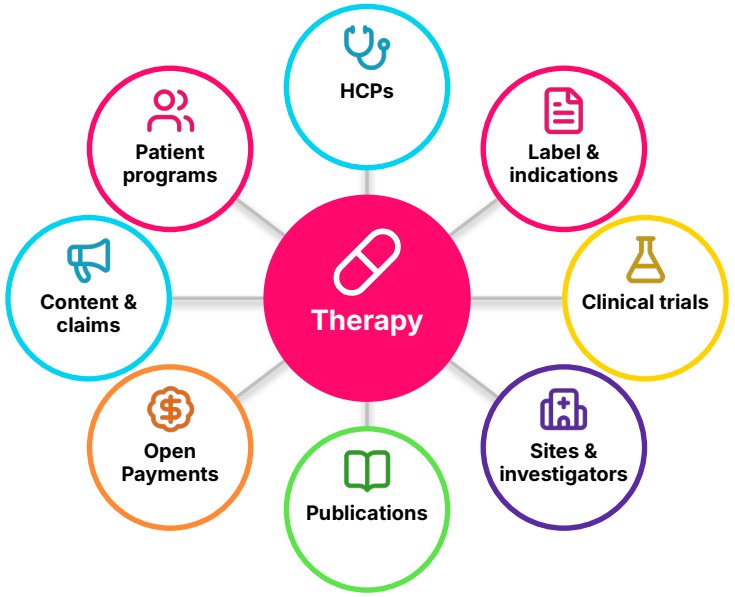
- **NPPES** — HCP identity (NPI)
- **CMS Open Payments** — Industry payments to HCPs
- **Medicare Part D** — Prescribing by NPI and drug
- **ClinicalTrials.gov** — Trials, criteria, sites
- **DailyMed labels** — Approved indications & claims
- **PubMed** — Publications & authors

Python ETL

- load_nppes
  load_open_payments
- load_partd
  load_labels
- ingest_trials
  load_pubmed
- resolve_npi
  embed_nodes

Knowledge graph

11 Entity types · **20** Relationships · **7** Vector indexes



Read & intelligence layer

Graph API
REST · GraphQL · MCP

Graph Assistant
agents · any LLM

- Pre-call briefing
- Next-best-action
- MLR pre-review
- Trial feasibility
- Patient matcher

PHI masking · guardrails in & out · human approval · audit trail

- **Life Sciences Cloud console**
- **Marketing Cloud**
- **Natural-language Q&A**

Key capabilities

HCP 360 · Next-best-action · MLR pre-review · Trial feasibility · Patient recruitment · GraphRAG Q&A · Agentic AI

VALUE & BENEFITS

More relevant HCP engagement, faster content approval and trials that enroll on time.

~40%

less trial screening time with NIH's TrialGPT approach

Source: NIH, Nature Communications 2024

↓

MLR cycle time per asset

Pilot target

↑

HCP engagement rate

Pilot target

$14.67B

in HCP payments in CMS Open Payments (PY2025), linkable by NPI

Source: CMS via IntuitionLabs

● Business

Faster launches, rep productivity, shorter trials

● Clinical

Accurate, on-label information; more patients reach trials

● IT & data

One therapy graph shared by commercial, medical and clinical

● Compliance

Sunshine Act and PhRMA Code guardrails built in

THE PROBLEM

Medication decisions depend on data scattered across many systems

Medication lists are messy, interaction alerts are noisy, prior auth is manual, adherence slips and shortages hit without warning. General-purpose AI models also miss dangerous interactions.

Who feels it: Health-system and specialty pharmacy, PBMs, pharma patient-services hubs, drug safety

PROBLEMS IT SOLVES

Messy medication lists
Reconciled by hand at admission and discharge

Missed interactions
Noisy alerts hide real risks

Prior-auth backlogs
Packets assembled from charts by hand

Poor adherence
Refill gaps go unnoticed

Missed adverse events
Safety reports lost in support calls

Shortages
Substitutes chosen under pressure

OUR SOLUTION

RxGraph

Pharmacy & Medication Intelligence Platform

One graph of every medication (ingredients, interactions, coverage, safety and supply) with agents that reconcile, check, authorize and substitute.

- **Medication reconciliation** free text to standard codes, discrepancies flagged
- **Interaction & PGx safety** graph lookup first; AI only explains
- **Benefits & prior auth** evidence mapped to payer criteria (CMS-0057-F ready)
- **Adherence & adverse events** refill nudges and safety-report capture
- **Shortage substitution** equivalent options checked against formulary

BUILT ON

Health Cloud | Service Cloud | Agentforce | EHR (FHIR)

RxGraph / Pharmacy & Medication Intelligence Platform

Connecting medications, interactions, coverage, safety and supply with a knowledge graph

Data sources

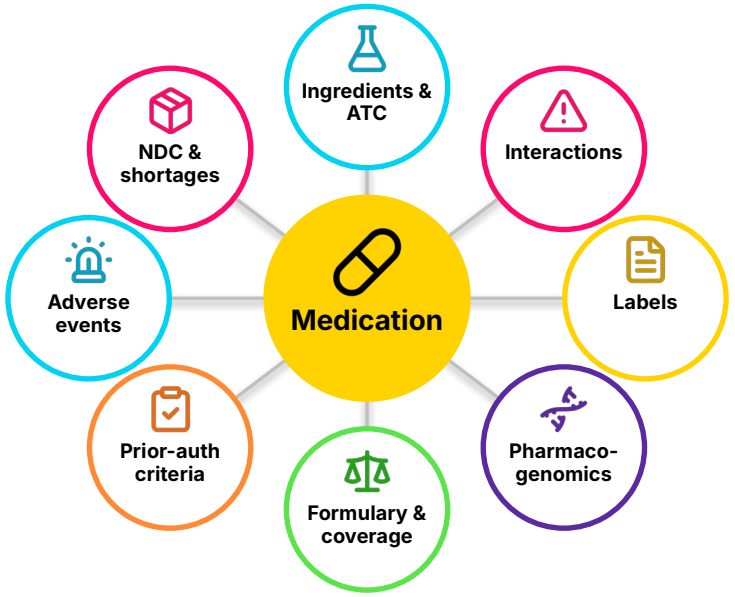
- **RxNorm** — Drug identity (RxCUI)
- **DailyMed / openFDA** — Labels, indications, warnings
- **openFDA FAERS** — Adverse event reports
- **CPIC / PharmGKB** — Gene–drug guidance
- **CMS Part D & LCD/NCD** — Formulary & coverage rules
- **FDA shortages & NDC** — Supply status

Python ETL

- load_rxnorm
  load_labels
- ingest_faers
  load_cpic
- load_formulary
  load_policies
- ingest_shortages
  embed_nodes

Knowledge graph

- **12** Entity types
- **22** Relationships
- **6** Vector indexes



Read & intelligence layer

Graph API
REST · GraphQL · MCP

Graph Assistant
agents · any LLM

- Reconciler
- Interaction checker
- Prior-auth builder
- Adherence coach
- Substitute finder

PHI masking · guardrails in & out · human approval · audit trail

- **Pharmacist worklist**
- **Health Cloud / EHR**
- **Natural-language Q&A**

Key capabilities

- Med reconciliation
- Interaction safety
- Prior auth
- Adherence
- Adverse events
- Shortage substitution
- GraphRAG Q&A

VALUE & BENEFITS

Safer medication lists, faster access to therapy and fewer surprises from shortages.

4 of 4

AI models tested missed a known interaction, so answers must be grounded

Source: ERCSP 2024

↓

Pharmacist minutes per reconciliation

Pilot target

↑

Prior-auth first-pass approval

Pilot target

Jan 2027

CMS prior-authorization API deadline (CMS-0057-F)

Source: CMS

● Business

Fewer readmissions, faster time to therapy, less waste

● Clinical

Fewer adverse drug events, continuity during shortages

● IT & data

Standard drug codes and FHIR-ready prior-auth APIs

● Compliance

Pharmacist approves every change; adverse events routed to safety

THE PROBLEM

Device data is everywhere, but nobody sees the whole picture

Remote-monitoring readings flood care teams, complaints pile up against tight FDA deadlines, recalls are hard to trace to patients, and connected devices carry known vulnerabilities.

Who feels it: MedTech quality & regulatory, virtual care / RPM programs, biomedical engineering

PROBLEMS IT SOLVES

Alert fatigue
Too many readings, too few real signals

Complaint backlogs
Manual triage and inconsistent coding

Late FDA reports
Reportability decided under time pressure

Recall traceability
Which patients have the affected device?

Device vulnerabilities
Known CVEs on connected devices

RPM billing missed
Data-day and minute rules tracked by hand

OUR SOLUTION

DeviceGraph

Medical Device & IoT Intelligence Platform

One graph that follows every device (UDI) from patient and telemetry to complaints, recalls and vulnerabilities, with agents for triage, reporting and remediation.

- **Remote-monitoring triage** rules and models decide; AI explains
- **Complaint & reportability** intake, IMDRF coding, FDA / EU MDR advice
- **Recall & UDI tracing** device to patient in one query
- **Post-market trends** early signals from MAUDE and internal data
- **Device cybersecurity** CVE-to-asset matching and patch plans

BUILT ON

Health Cloud | Service Cloud | MuleSoft | Tableau

DeviceGraph / Medical Device & IoT Intelligence Platform

Connecting devices, patients, telemetry, complaints, recalls and security with a knowledge graph

Data sources

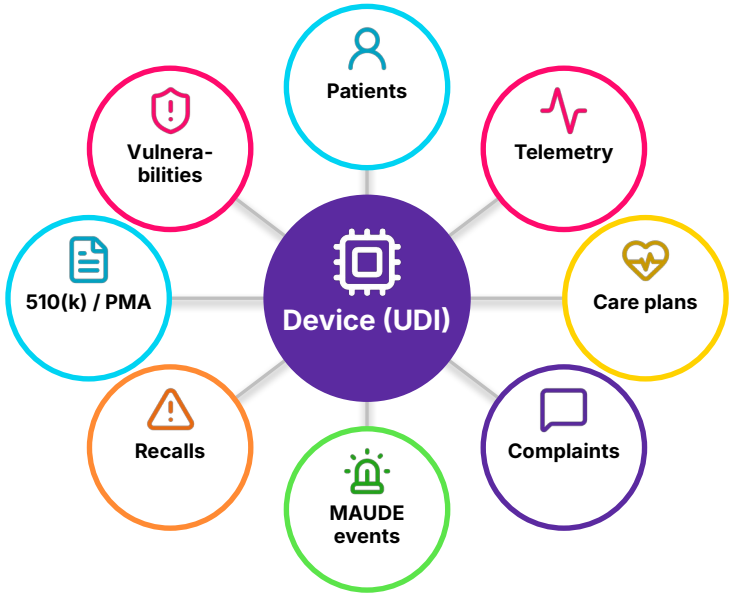
- **Device telemetry** — Wearables & home devices (FHIR)
- **FDA GUDID** — UDI device identity
- **FDA MAUDE** — Device adverse events
- **FDA recalls & 510(k)** — Recalls, clearances, predicates
- **CISA ICSMA & NVD** — Device vulnerabilities
- **Complaints & service** — Service Cloud cases

Python ETL

- ingest_telemetry to_fhir
- load_gudid ingest_maude
- load_recalls load_510k
- load_cves embed_nodes

Knowledge graph

12 Entity types · **21** Relationships · **5** Vector indexes



Read & intelligence layer

Graph API
REST · GraphQL · MCP

Graph Assistant
agents · any LLM

- RPM triage
- Complaint intake
- Reportability advisor
- Trend detector
- Patch planner

PHI masking · guardrails in & out · human approval · audit trail

- **Care team console**
- **Quality & regulatory**
- **Natural-language Q&A**

Key capabilities

Remote monitoring · Complaint handling · FDA reporting · Recall tracing · Cybersecurity · GraphRAG Q&A · Agentic AI

VALUE & BENEFITS

Fewer, better alerts, faster complaint handling and devices you can trace and secure.

2M+

medical device reports FDA receives each year (MAUDE)

Source: FDA

↓

Alerts per nurse per day

Pilot target

↓

Complaint cycle time and late FDA reports

Pilot target

↓

Time to remediate critical device vulnerabilities

Pilot target

Business

Billable virtual care, lower cost of quality, fewer surprise recalls

Clinical

Earlier intervention and safety issues found sooner

IT & data

Standards-based device integration (FHIR, UDI)

Compliance

21 CFR 803, EU MDR vigilance; people make every reportability call

THE PROBLEM

From supplier to patient, the chain is only visible in pieces

Supplier risks are hidden, deviation investigations take weeks, repeat findings lead to FDA 483s, and shortages are discovered too late to plan around.

Who feels it: Pharma & biotech manufacturing, CDMOs, quality, supply chain, distributors

PROBLEMS IT SOLVES

Hidden supplier risk
Single-source materials and facilities

Slow investigations
Deviations start from a blank page

Repeat deviations
Root causes never connected; CAPAs overdue

Batch review bottleneck
eBMR review delays release

Inspection findings
Repeat FDA 483 themes

Late shortage warning
Stock-outs discovered too late

OUR SOLUTION

ChainGraph

Manufacturing & Supply Chain Intelligence Platform

One graph of the product chain (supplier, material, plant, batch, distribution) with agents for quality, risk and shortage planning.

- **Supplier & material risk** single-source and facility risk map
- **Deviation & CAPA copilot** similar events, root cause and CAPA drafts
- **Batch-record review** exceptions flagged before QA release
- **Inspection readiness** matches findings to FDA 483 themes
- **Shortage early warning** supply signals and substitution plans

BUILT ON

Quality system (e.g. Veeva Vault QMS) | MES / LIMS / ERP | Tableau | Any cloud

ChainGraph / Manufacturing & Supply Chain Intelligence Platform

Connecting suppliers, plants, batches, quality and distribution with a knowledge graph

Data sources

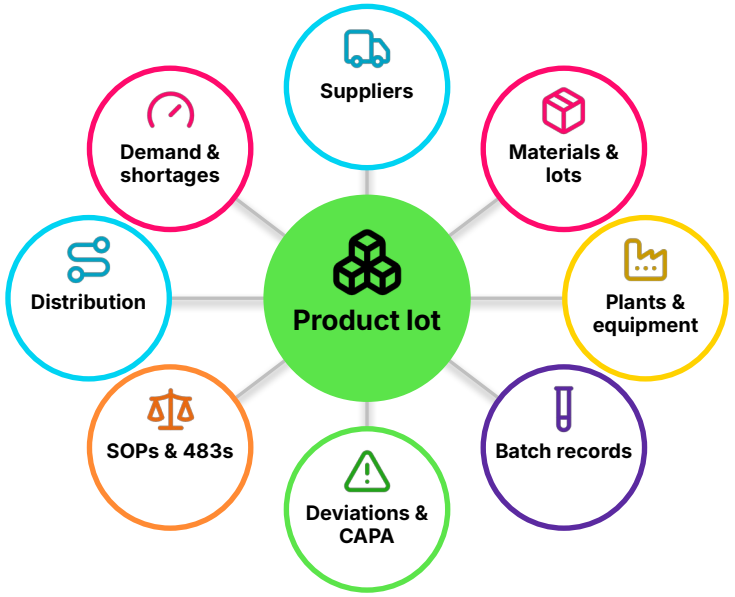
- **ERP & suppliers** — Suppliers, materials, orders
- **MES / eBMR & LIMS** — Batches, equipment, tests
- **Quality system (QMS)** — Deviations, CAPAs, changes
- **FDA 483s & letters** — Inspection findings
- **FDA shortages & recalls** — Market supply signals (NDC)
- **DSCSA / distribution** — Serialization & inventory

Python ETL

- load_erp
  load_suppliers
- ingest_mes
  ingest_lims
- load_qms
  load_483s
- ingest_shortages
  embed_nodes

Knowledge graph

13 Entity types · **24** Relationships · **6** Vector indexes



Read & intelligence layer

Graph API
REST · GraphQL · MCP

Graph Assistant
agents · any LLM

- Supplier risk
- Deviation triage
- Root-cause drafter
- CAPA checker
- Shortage early warning

PHI masking · guardrails in & out · human approval · audit trail

- **Quality workbench**
- **Supply dashboard**
- **Natural-language Q&A**

Key capabilities

Supplier risk · Deviation & CAPA · Batch review · Inspection readiness · Shortage warning · GraphRAG Q&A · Agentic AI

VALUE & BENEFITS

Close investigations faster, stop repeat deviations and see shortages coming, with QA always in control.

40%

faster deviation closure reported in industry GenAI case studies

Source: McKinsey (vendor-reported)

323

active US drug shortages at the Q1 2024 peak

Source: ASHP

↓

Repeat deviations and overdue CAPAs

Pilot target

↑

Days of warning before a stock-out

Pilot target

Business

Right-first-time batches, less lost product, less emergency buying

Clinical

Consistent quality and continuity of supply for patients

IT & data

Validated, versioned pipelines (GAMP 5 / CSA)

Compliance

QA approves every AI draft; Part 11 audit trail

How we engage: from idea to production

Ideation workshop
2 WEEKS

Pick the graph and first modules, map data and stakeholders, agree success metrics.

Proof of concept
6–8 WEEKS

Working graph and agents on public and synthetic data, architecture signed off, sponsor demo.

Pilot
~3 MONTHS

Real users, governed client data, measured against the agreed KPIs.

Scale
ONGOING

Add modules and sites, then the next graph on the same platform.

Next step: pick one graph and run a 2-week ideation workshop with the client.

Sales cheat sheet: which graph to bring

Bring	Listen for...	Who to talk to
CareGraph Care Engagement	Care gaps, low screening rates, no-shows, outreach that doesn't convert, network leakage, price questions	CMO / CNO, VP Population Health, Patient Access, Marketing
TherapyGraph Life Sciences	Launch planning, HCP engagement, slow MLR review, trial enrollment delays, site selection	Brand / commercial leads, Medical Affairs, Clinical Operations
RxGraph Pharmacy & Drug	Medication errors, noisy alerts, prior-auth backlogs, CMS-0057-F, adherence, adverse events, shortages	Chief Pharmacy Officer, VP Patient Services, Drug Safety
DeviceGraph Medical Devices & IoT	Remote monitoring programs, alert fatigue, complaint backlogs, FDA / EU MDR reporting, recalls, device security	VP Quality / Regulatory, Post-Market Surveillance, Virtual Care, Biomed / CISO
ChainGraph Manufacturing & Supply Chain	Deviation backlogs, FDA 483s / warning letters, overdue CAPAs, supplier risk, drug shortages	VP Quality / QA, Site Heads, Supply Chain, Hospital Pharmacy Directors



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